

EXHIBIT 35

A Model for the Trench Transistor

SANJAY BANERJEE, MEMBER, IEEE, AND D. MARK BORDELON, MEMBER, IEEE

Abstract—This paper discusses a model for the trench transistor used in the trench transistor cell (TTC), which is employed in Texas Instruments' 4-Mbit DRAM. PISCES-II simulations are used to study the unique characteristics that result from the nonuniform doping along the channel and the nonuniform gate oxides in these transistors. The simulations are correlated with experimental data and an analytical description is proposed to qualitatively explain the observed behavior.

I. INTRODUCTION

THE QUEST for higher packing densities in IC's has led to 3-D integration involving silicon-on-insulator schemes or trench-based technologies. Texas Instruments' 4-Mbit DRAM, using a trench transistor cell (TTC), is an example of the latter approach [1]. It uses a one-transistor cell where the pass transistor and the storage capacitor are both incorporated in a trench. Since these trench-based approaches involve making active MOS transistors on the sidewalls of a trench, they have certain unique features compared to planar MOSFET's. The threshold adjust implant for these vertical MOSFET's results in a nonuniform doping profile along the channel, while the doping density is constant normal to the Si-SiO₂ interface. In contrast, for planar MOSFET's, while the doping may vary with depth in the substrate, it is uniform along the channel. Secondly, the trench sidewalls, typically, have a variety of crystallographic orientations. Hence while planar MOSFET's are generally made on {100} silicon with uniform gate oxide thickness and interface properties, trench transistors, made on the trench sidewalls, have a variation of gate-oxide thickness, fixed charge, and fast state density in the channel region.

This paper will address the modeling of MOSFET's with nonuniform doping along the channel and nonuniform gate oxides, using the 4-Mbit DRAM TTC as a vehicle.

II. CELL STRUCTURE

An 8- μm -deep trench ($1.3\ \mu\text{m} \times 1.5\ \mu\text{m}$) is etched using reactive ion etching in a 4.5- μm epitaxial layer on a p^+ substrate. A vertical pass transistor is made in the top 2 μm of the trench. The storage capacitor is also in the trench, below the pass transistor (Fig. 1(a)). The inner

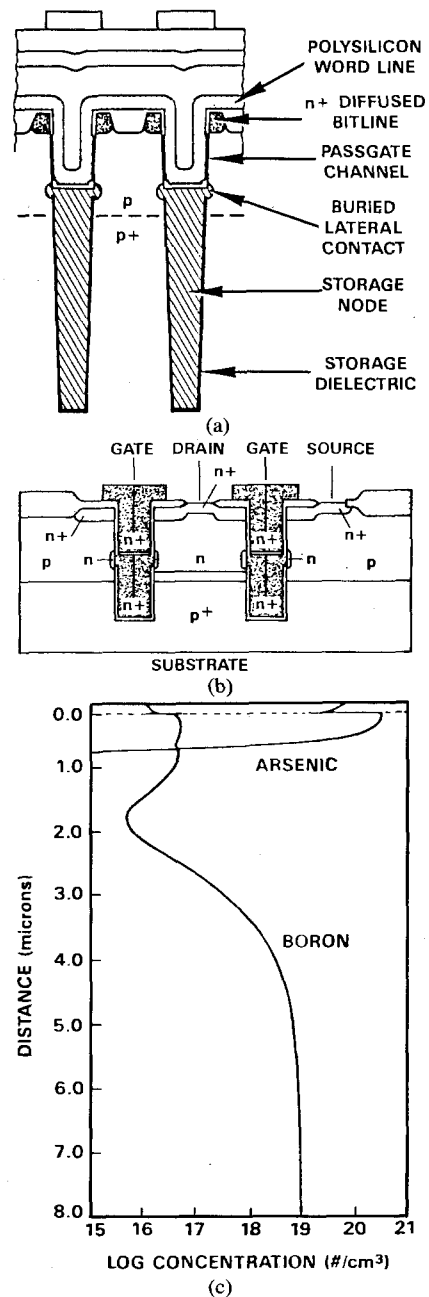


Fig. 1. (a) TTC schematic diagram. (b) Trench transistor test structure allowing access to the lateral contact (source) of the transistor. (c) SUPREM-III generated doping profile along the channel of the trench transistor.

plate of the capacitor is an n^+ polysilicon plug, and the outer plate is the p^+ substrate. The source of the transistor is connected to the n^+ polysilicon plug of the capacitor

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S. Banerjee was with the Semiconductor Process and Design Center, Texas Instruments Incorporated, Dallas, TX 75265. He is now with the Department of Electrical and Computer Engineering, University of Texas, Austin, TX 78712-1084.

D. M. Bordelon is with the Semiconductor Process and Design Center, Texas Instruments Incorporated, Dallas, TX 75265.

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by a lateral contact that is made by an oxide undercut etch and refilling with polysilicon [2], [3]. The drain, gate, and source of the trench pass transistor are made of diffused-buried- n^+ bitline, n^+ polysilicon wordline, and a lateral contact, respectively. The gate-oxide thickness and channel length are around 25 nm and 1.5 μm , respectively, while the transistor width is determined by the perimeter of the trench.

A "0" level is stored as 0 V and the "1" level as 5 V. The bitlines are precharged to $1/2 V_{cc}$ or 2.5 V and the substrate is held at -2.5 V. The bootstrapped wordlines are switched between 0 and 7 V to transfer data back and forth from the cell to the bitline.

Since the TTC structure does not allow direct access to the lateral contact (source) of the transistor, a special octagonal annular trench transistor was fabricated (Fig. 1(b)) that yields a transistor on the outer periphery in series with another in the center. The inner transistor is shorted by an n-tank diffusion and contacts to gate, drain, and source are obtained. The experimental characterization was done on this structure.

III. DEVICE SIMULATION

The trench transistor has been studied using a two-dimensional device simulator (PISCES-II) [4]. The doping profiles in the buried- n^+ bitline, lateral contact, and implanted channel region are simulated by SUPREM III (Fig. 1(c)) and provided as an input to PISCES-II where concentration-dependent and field-dependent mobility models have been used. The linear region and subthreshold transfer characteristics (Fig. 2(a)) indicate that a low threshold voltage V_t of 0.37 V (0.90 V) and good subthreshold slope S of 87 mV/dec (76 mV/dec) are achieved at a substrate bias of 0 V (-2.5 V), respectively. There is good agreement between measured and simulated values except that the observed soft turn-on behavior is not reproduced by the simulation. The reason for this will be discussed later. A fixed charge density of $5 \times 10^{10} \text{ cm}^{-2}$ and a fast state density of $5 \times 10^{10} \text{ cm}^{-2} \cdot \text{eV}^{-1}$ is used for the simulation. The channel mobility is estimated to be $480 \text{ cm}^2/\text{V} \cdot \text{s}$ from the data, but the theoretical gain K ($24.4 \mu\text{A}/\text{V}^2$) is degraded by a factor of about 2 because of the high source series resistance inherent in the test structure (estimated to be 200Ω) [5]. Since an external lumped series resistance cannot be used in PISCES-II, SPICE parameters for the intrinsic device were extracted from the PISCES-II simulation and used in a SPICE simulation with an external lumped source resistance of 200Ω to obtain the output characteristics. Experimental and simulated output characteristics shown in Fig. 2(b) indicate that good agreement (within 10 percent) is obtained in both the linear and the saturation regions.

IV. ANALYTICAL MODEL

For comparison, a transistor having a fixed gate oxide of 25 nm and an uniform channel doping equal to the peak concentration in the trench transistor above has a V_t of

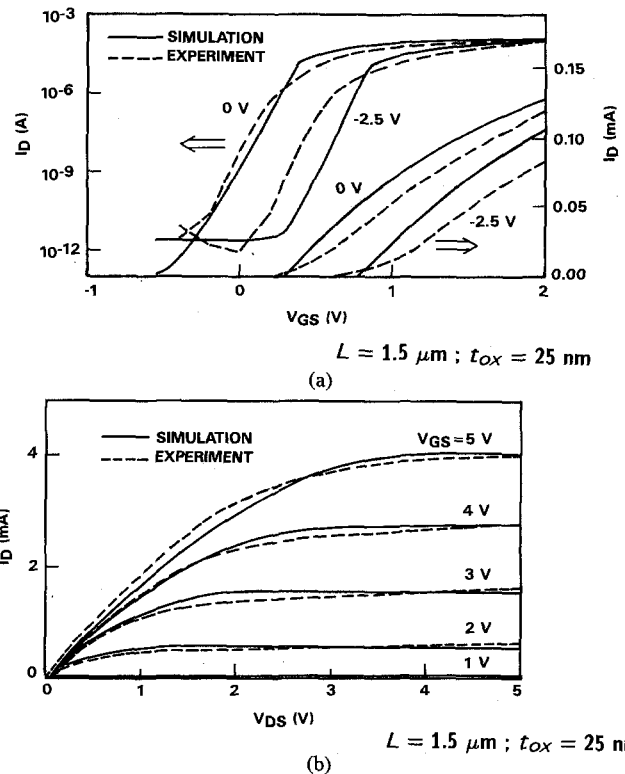


Fig. 2. (a) Linear and subthreshold characteristics of trench transistor. The PISCES-II simulations assume a nominal oxide thickness of 25 nm. (b) Experimental and simulated output characteristics of a $42\text{-}\mu\text{m}/1.5\text{-}\mu\text{m}$ trench transistor with a substrate bias of 0 V.

0.61 V (1.35 V) and an S of 104 mV/decade (77 mV/decade) at a substrate bias of 0 V (-2.5 V), according to simulation. This indicates that the V_t and subthreshold behavior are not simply determined by the peak channel doping, as would be predicted by a simple 1-D solution of Poisson's equation. The doping peak gives rise to a 2-D potential distribution.

A typical PISCES-II simulation of the equipotential contours in the trench transistor at inversion is shown in Fig. 3 for the doping profile of Fig. 1(c). The depletion region is compressed near the doping peak, and the equipotential patterns indicate that there is a longitudinal electric field component in addition to the transverse field because of the doping variation along the channel.

The nonuniform doping profile in the channel of the trench transistor may be approximated as

$$N(x, y) = N_a + N_p \exp \left[-\frac{(x - R)^2}{2\sigma^2} \right] \quad (1)$$

where N_a reflects the uniform (tank) doping and the Gaussian component is because of the V_t adjust implant (Fig. 1(c)). Assuming complete depletion in the depletion region in the channel, the potential ψ must satisfy the 2-D Poisson equation

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = \frac{q}{\epsilon_{\text{si}}} \left[N_a + N_p \exp \left[-\frac{(x - R)^2}{2\sigma^2} \right] \right] \quad (2)$$

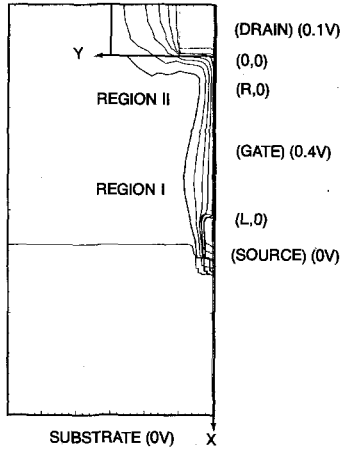


Fig. 3. PISCES-II simulation of equipotential contours and depletion edges in trench transistor. The potential contours are bunched closer together near the doping peak at $x = R$. The electric field has an x and y component in region II but just a y -component in region I.

Far from the doping peak (region I), where the doping is constant (N_a), the gradual channel approximation may be applied, i.e., there is no longitudinal component of the electric field (Fig. 3). Equation (2) then simplifies to a 1-D Poisson equation

$$\frac{\partial^2 \psi}{\partial y^2} = \frac{qN_a}{\epsilon_{si}} \quad (3)$$

which gives the potential distribution for an uniformly doped long-channel MOSFET.

Equation (3) is solved subject to the boundary conditions

$$\psi(0) = \psi_s(I) \quad (3a)$$

and

$$\psi(d) = 0 \quad (3b)$$

where d is the depletion width and $\psi_s = (= qN_a d^2 / 2\epsilon_{si})$ is the surface potential. Then

$$\psi(y) = \frac{qN_a}{\epsilon_{si}} \left[\frac{(d-y)^2}{2} - \frac{d^2}{2} \right] + \psi_s. \quad (4)$$

When the gate is biased ($V_g = V_{t1}$) such that the surface potential bandbending is $2\phi_F$ for a doping of N_a , the Gaussian doping creates an additional surface potential barrier (in region II) which is approximately given by

$$\begin{aligned} V_{t1} &= V_{FB} + 2\phi_F + \frac{1}{C_{ox}} \sqrt{(4\epsilon_{si} q \phi_F N_a)} \\ &= V_{FB} + 2\phi_F - \Delta\psi(x) + \frac{1}{C_{ox}} \cdot \sqrt{\left(2\epsilon_{si} q (2\phi_F - \Delta\psi(x)) \left(N_a + N_p \exp \left[-\frac{(x-R)^2}{2\sigma^2} \right] \right) \right)}. \end{aligned} \quad (5)$$

Therefore

$$\Delta\psi(x) \cong \frac{kT}{q} \frac{N_p}{N_e} \exp \left[-\frac{(x-R)^2}{2\sigma^2} \right] \quad (6)$$

where

$$N_e = N_a \left[2 \frac{kT}{q} \left(\frac{4C_{ox} \phi_F + \sqrt{4\phi_F \epsilon_{si} q N_a}}{\sqrt{(4\phi_F)^3 \epsilon_{si} q N_a}} \right) \right].$$

Therefore, when $V_g = V_{t1}$ such that the channel is inverted ($\psi_s = 2\phi_F$) in region I, the more heavily doped region near the gaussian peak remains noninverted (region II) (Fig. 3).

Assuming that at $V_g = V_{t1}$ a doping of $(N_a + N_s)$ corresponds to a surface potential of $(2\phi_F - 3kT/q)$ leading to depletion

$$\begin{aligned} \frac{3kT}{q} C_{ox} &= \sqrt{(4\epsilon_{si} q \phi_F N_a)} \sqrt{\left(1 + \frac{N_s}{N_a} \right)} \\ &\cdot \sqrt{\left(1 - \frac{3kT}{2q\phi_F} \right) - \sqrt{(4\epsilon_{si} q \phi_F N_a)}} \\ &\cong \sqrt{(4\epsilon_{si} q \phi_F N_a)} \left[\frac{N_s}{2N_a} - \frac{3kT}{4q\phi_F} \right] \end{aligned} \quad (7)$$

provided $N_s < N_a$.

For a Gaussian profile, at $x = R + / - S$

$$N_s = N_p \exp \left[-\frac{S^2}{2\sigma^2} \right]. \quad (8)$$

Therefore

$$S = \sqrt{2} \sigma$$

$$\sqrt{\left[\ln \left(\frac{N_p}{2N_a} \right) - \ln \left(\frac{\frac{3kT}{q} C_{ox}}{\sqrt{4\epsilon_{si} q \phi_F N_a}} + \frac{3kT}{4q\phi_F} \right) \right]}. \quad (9)$$

Substituting appropriate values for the trench transistor, we obtain $S \cong 2\sigma$.

The trench transistor may be considered to be two transistors in series, one corresponding to the Gaussian doping region of length $2S$, and the other corresponding to the doping N_a of length $(L - 2S)$, where L is the physical channel length. This explains why the linear region characteristics of the trench transistor show two slopes (Fig. 4(a)). For $V_g > V_{t1}$, region II is noninverted while region I goes into inversion (Fig. 3). For V_g slightly above V_{t1} , region II is weakly inverted while region I is so strongly

inverted that it may be considered to be an extension of source-drain (Fig. 4(b)). For this gate bias range, the gain is proportional to $1/(2S)$. For much higher gate biases, however, the entire channel is strongly inverted and the gain varies as $1/L$. This nonlinearity is not significant if

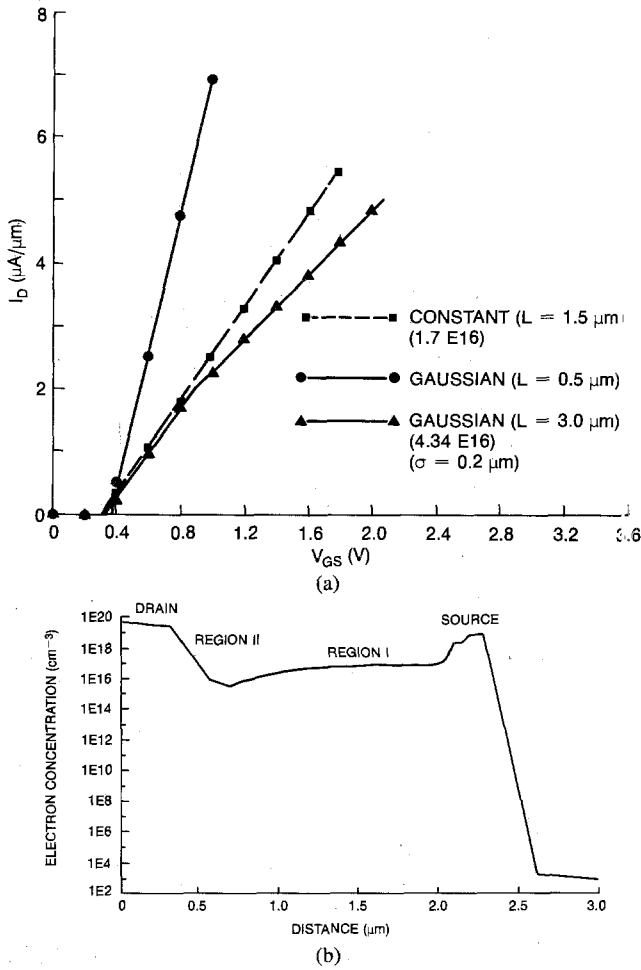


Fig. 4. (a) PISCES-II simulation of linear region characteristics for trench transistor and uniformly doped transistor. (b) Electron concentration along the surface at inversion for trench transistor in Fig. 3, showing the high concentration in region I, and the lower concentration in region II.

L is comparable to $(2S)$ because, then, it is not possible to distinguish two distinct transistors in series. Also, this effect is not seen for uniform doping, proving that the nonlinearity is related to the nonuniform doping along the channel (Fig. 4(a)).

To study the dependence of the trench transistor on the doping parameters N_a , N_p , and σ , the additional barrier due to the Gaussian component is calculated as a perturbation of the surface potential profile due to N_a alone. We have

$$V_g = V_{ox} + V_{FB} + \psi_s \quad (10)$$

where

$$V_{ox} = t_{ox} E_{ox} = t_{ox} \frac{\epsilon_{si}}{\epsilon_{ox}} E_{si} = \frac{\epsilon_{si}}{C_{ox}} \left(-\frac{\partial \psi}{\partial y} \right). \quad (11)$$

From (6) and (10), the difference in the y component of the surface electric field between regions I and II is

$$\Delta E_{ys} = \frac{C_{ox}}{\epsilon_{si}} \frac{kT}{q} \frac{N_p}{N_e} \exp \left[-\frac{(x-R)^2}{2\sigma^2} \right]. \quad (12)$$

From (4), the depletion width is approximately

$$d \cong \sqrt{\frac{\left(4\epsilon_{si} \frac{kT}{q} \ln \frac{N_a}{n_i} \right)}{qN_a}}. \quad (13)$$

Therefore

$$\frac{\partial^2 \psi(II)}{\partial y^2} - \frac{\partial^2 \psi(I)}{\partial y^2} = \frac{C_{ox}}{\epsilon_{si} d} \frac{kT}{q} \frac{N_p}{N_e} \exp \left[-\frac{(x-R)^2}{2\sigma^2} \right]. \quad (14)$$

Now, to obtain the additional potential barrier in region II, the Poisson equation is applied at $V_{t1} < V_g < V_{t2}$, such that region I is inverted, but region II is not. In region I

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \cong \frac{\partial^2 \psi}{\partial y^2} \cong \frac{q}{\epsilon_{si}} [N_a + n_{inv}] \quad (15)$$

where n_{inv} is the average inversion-layer charge density in the inversion layer. The inversion-layer areal charge density is

$$Q_n = C_{ox}(V_g - V_{t1}). \quad (16a)$$

Therefore

$$n_{inv} = \frac{C_{ox}(V_g - V_{t1})}{x_{eff}} \quad (16b)$$

where x_{eff} , the inversion layer thickness, is determined by a self-consistent solution of Poisson's equation coupled with Schrodinger's equation [6]. x_{eff} decreases with increasing N_a , but is not very sensitive to V_g .

In region II

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = \frac{q}{\epsilon_{si}} \left[N_a + N_p \exp \left[-\frac{(x-R)^2}{2\sigma^2} \right] \right]. \quad (17)$$

From (14), (15), and (17), in region II

$$\begin{aligned} \frac{\partial^2 \psi}{\partial x^2} &\cong \frac{q}{\epsilon_{si}} \left[\left[N_p \left(1 - \frac{C_{ox}}{qdN_e} \frac{kT}{q} \right) \right] \right. \\ &\quad \cdot \exp \left[-\frac{(x-R)^2}{2\sigma^2} \right] - n_{inv} \left. \right] \\ &= \frac{q}{\epsilon_{si}} \left[N_1 \exp \left[-\frac{(x-R)^2}{2\sigma^2} \right] - n_{inv} \right] \end{aligned} \quad (18)$$

where

$$N_1 = N_p \left[1 - \left(\frac{C_{ox}}{qdN_e} \frac{kT}{q} \right) \right]. \quad (18a)$$

Therefore, by the above technique, the 2-D Poisson equation in region II has been decoupled into two 1-D equations along x and y . The surface potential in region II (and hence the transistor behavior) may be obtained by solving (18) subject to the boundary conditions

$$\psi(R \pm S) = \psi_s \quad (18b)$$

and

$$\frac{\partial \psi(R + / - S)}{\partial x} = 0. \quad (18c)$$

Then

$$\begin{aligned} \frac{\partial \psi}{\partial x} = & \frac{qN_1\sqrt{\pi}}{2\epsilon_{si}} \left[\frac{2}{\sqrt{\pi}} \int_{R-S}^x \exp \left[-\frac{(x-R)^2}{2\sigma^2} \right] dx \right] \\ & - \frac{qn_{inv}}{\epsilon_{si}} [x - (R - S)]. \end{aligned} \quad (19a)$$

Therefore

$$\begin{aligned} \frac{\partial \psi}{\partial x} = & \frac{qN_1\sigma\sqrt{\pi/2}}{\epsilon_{si}} \left[2 - \operatorname{erfc} \left[\frac{(x-R)}{\sqrt{2}\sigma} \right] \right. \\ & \left. - \operatorname{erfc} \left[\frac{S}{\sqrt{2}\sigma} \right] \right] - \frac{qn_{inv}}{\epsilon_{si}} [x - (R - S)]. \end{aligned} \quad (19b)$$

Then

$$\begin{aligned} \psi = & \frac{qN_1\sigma\sqrt{\pi/2}}{\epsilon_{si}} \int_{R-S}^x \left[\operatorname{erfc} \left[\frac{(x-R)}{\sqrt{2}\sigma} \right] \right] dx \\ & + \frac{qN_1\sigma\sqrt{\pi/2}}{\epsilon_{si}} \left[2 - \operatorname{erfc} \left[\frac{S}{\sqrt{2}\sigma} \right] \right] [x - (R - S)] \\ & - \frac{qn_{inv}}{2\epsilon_{si}} [x - (R - S)]^2 + \psi_s(I). \end{aligned} \quad (20)$$

To integrate the first term, we use the identities [7]

$$\int_z^\infty \operatorname{erfc}(t) dt = -(z) \operatorname{erfc}(z) + \frac{1}{\sqrt{\pi}} \exp[-z^2] \quad (21a)$$

and

$$\int_0^\infty \operatorname{erfc}(t) dt = \frac{1}{2\Gamma(3/2)} = \frac{1}{\sqrt{\pi}}. \quad (21b)$$

Then

$$\begin{aligned} \psi = & -\frac{qN_1\sqrt{\pi}\sigma^2}{\epsilon_{si}} \left[\frac{S}{\sqrt{2}\sigma} \left(2 - \operatorname{erfc} \left(\frac{S}{\sqrt{2}\sigma} \right) \right) \right. \\ & + \left(\frac{x-R}{\sqrt{2}\sigma} \right) \operatorname{erfc} \left(\frac{x-R}{\sqrt{2}\sigma} \right) \\ & + \frac{1}{\sqrt{\pi}} \left(\exp \left(-\frac{S^2}{2\sigma^2} \right) - \exp \left(-\frac{(x-R)^2}{2\sigma^2} \right) \right) \Big] \\ & + \frac{qN_1\sqrt{\pi}\sigma}{\sqrt{2}\epsilon_{si}} \left[2 - \operatorname{erfc} \left(\frac{S}{\sqrt{2}\sigma} \right) \right] (x - (R - S)) \\ & - \frac{qn_{inv}}{2\epsilon_{si}} (x - (R - S))^2 + \psi_s(I). \end{aligned} \quad (22)$$

The maximum barrier height is at the doping peak at $x = R$.

$$\begin{aligned} \psi_b = & \psi(R) - \psi_s(I) \\ = & \frac{qN_1\sigma^2}{\epsilon_{si}} \left(1 - \exp \left[-\frac{S^2}{2\sigma^2} \right] \right) - \frac{qn_{inv}S^2}{2\epsilon_{si}}. \end{aligned} \quad (23)$$

At $V_g = V_{t1}$, which is the threshold for inversion in region I, $n_{inv} = 0$. Also $S = 2\sigma$ for typical trench transistor parameters.

Therefore, the peak barrier height in the channel (at near-flat-band bias) is approximately

$$\psi_b \cong \frac{0.135q}{\epsilon_{si}} \left[1 - \left(\frac{C_{ox}}{qN_e} \frac{kT}{q} \right) \right] N_p \sigma^2 \quad (24)$$

provided $N_p < N_a$. From (24), it is seen that the transistor turn-on and subthreshold behavior are determined by the peak doping ($N_a + N_p$), reduced due to the two-dimensionality of the equipotential contours, by a factor that is controlled by, among other things, the σ of the distribution.

V. PARAMETRIC DEPENDENCE

Two process parameters that can be varied most easily in the trench transistor are the V_t implant, which controls the doping profile along the channel, and the etchback of the trench, which controls the length of the vertical transistor.

Results of PISCES-II simulations for a 280-keV boron V_t implant of various doses are shown in Fig. 5(a) as a function of peak concentration ($N_a + N_p$). Good agreement is obtained for parameters such as V_t and body effect between simulation and experimental data. Since the cell V_t implant energy and thermal schedule are constant, the standard deviation σ is approximately constant for all data points.

$$\sigma^2 = \sigma_0^2 + 2Dt \quad (25)$$

where “ $2\sigma_0$ ” is the straggle of the as-implanted profile and “ $2Dt$ ” is the broadening of the profile because of thermal diffusion. N_a is constant ($1E16 \text{ cm}^{-3}$), but N_p , and hence the barrier height ψ_b increases with dose. This causes V_t and body effect to increase with implant dose.

The simulations for a transistor with uniform doping equal to the peak doping in the trench transistor result in a higher value for V_t and body effect than for the Gaussian profile, thereby confirming that the barrier height ψ_b is lower than that given by a constant doping of ($N_a + N_p$) because of the 2-D potential profile.

Similar observations apply to the variation of S with peak doping. S degrades with higher dose because the depletion capacitance in the channel increases and reduces the effectiveness of the gate in modulating the surface potential (Fig. 5(b)).

Since the previous study varied N_p , but kept σ approximately the same, the effect of varying σ while keeping ($N_a + N_p$) fixed was studied using PISCES-II (Fig. 6). A simple 1-D model would have predicted that V_t is kept

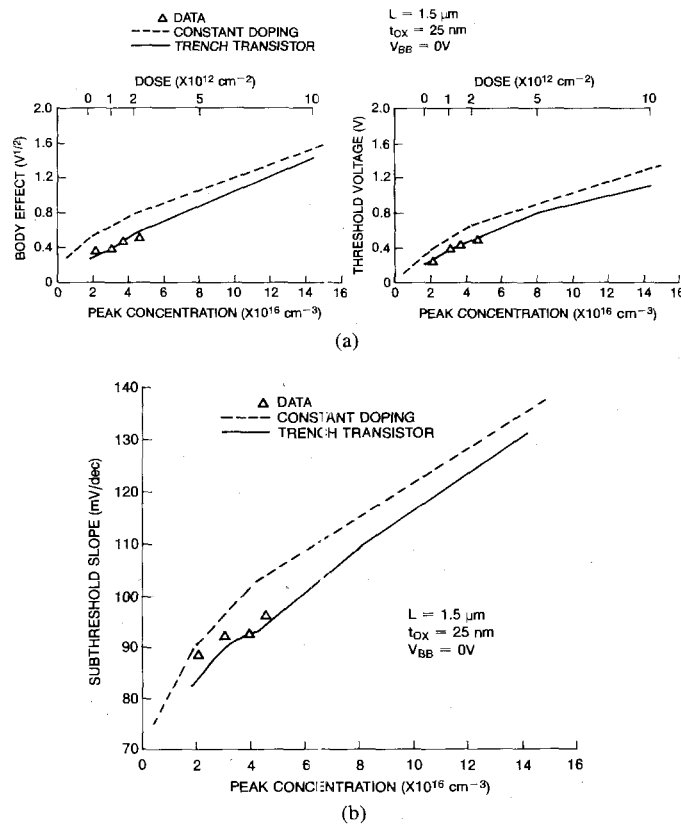


Fig. 5. (a) Threshold and body effect variation with cell V_t implant dose. (b) Subthreshold slope variation with dose.

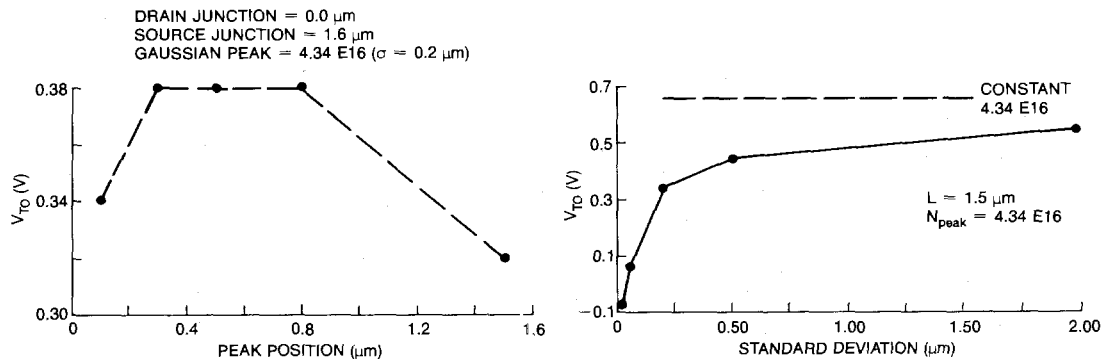


Fig. 6. PISCES-II simulation of V_t variation with the σ of the Gaussian doping profile and the peak position R of the profile.

constant because the peak doping is kept constant. However, ψ_b decreases with σ according to (25). PISCES-II simulations confirm that as the σ of the Gaussian profile is reduced, the effective barrier and V_t drops to the value determined by N_a ($1\text{E}15 \text{ cm}^{-3}$ in these simulations). As σ increases, V_t asymptotically approaches the value for an uniform doping of ($N_a + N_p$).

The variation of V_t and S with L as the lateral contact position is varied is shown in Fig. 7. Good agreement is obtained between simulated and measured values of S if a fast-state density of $2\text{E}11 \text{ cm}^{-2} \cdot \text{eV}^{-1}$ is assumed. The V_t of the trench transistor is seen to be constant even for very short channel lengths according to simulation and data. In contrast, for a uniformly doped channel, short-channel effects such as V_t lowering are quite pronounced.

In the uniformly doped case, as L reduces, a larger fraction of the bulk depletion charge under the gate is associated with the source-drain; this charge sharing leads to V_t lowering. Comparison of the equipotential contours for a $0.5\text{-}\mu\text{m}$ -long MOSFET with uniform doping ($1.7\text{E}16 \text{ cm}^{-3}$) and a trench transistor show that a larger part of the depletion charge is associated with the source-drain for the uniformly doped case. In the trench transistor, most of the depletion charge is near the peak of the Gaussian profile and is spatially separated from the source-drain, thereby minimizing short-channel V_t lowering effects. This is an advantage in the context of using the trench transistor as a pass transistor in the 4-Mbit DRAM because the V_t is insensitive to trench etchback (and hence, transistor length) variations. V_t and S increase for $L = 2$

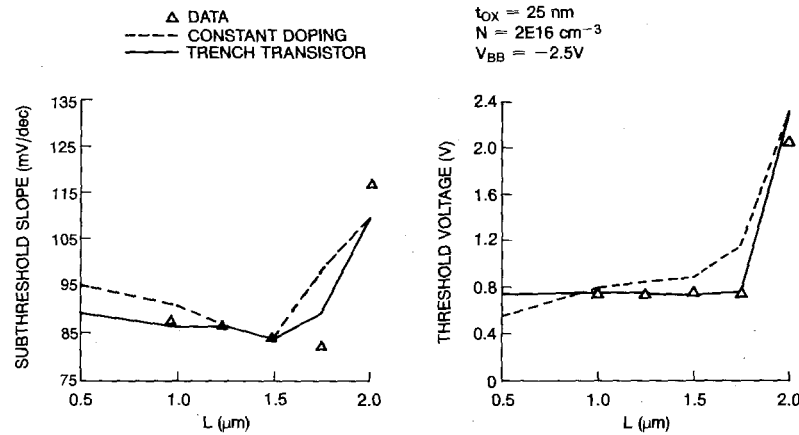


Fig. 7. V_t and subthreshold slope variation with channel length as the lateral contact position is moved up or down.

μm because the undiffused boron from the p^+ substrate increases the channel doping in the vicinity of the lateral contact.

The barrier height ψ_b is independent of the peak position R (see (24)). PISCES-II simulations confirm that V_t is independent of R (Fig. 6) unless the peak is very close to the source-drain, in which case there is some V_t lowering because some of the depletion charge is “shared” with the source-drain. Since ψ_b is independent of R , the trench transistor characteristics are not asymmetric when the source-drain are interchanged. For the trench transistor used in the 4-Mbit DRAM, PISCES-II simulations indicate only a 2-percent difference in saturation drain current at $V_{gs} = V_{ds} = 5$ V when source-drain are interchanged. Experimentally, however, there is a greater asymmetry because of source-drain series resistance effects.

The gain of the trench transistor is proportional to $1/L$ for gate biases much above V_t where regions I and II are both strongly inverted. Simulation of the trench transistor and for devices with constant doping show that the gain varies as $1/L$. Experimental values of gain are degraded because of source-drain series resistance effects (Fig. 8).

VI. ORIENTATION DEPENDENCE

Although the trenches are drawn rectangular, they have an oval cross section after patterning and reactive-ion etching, and therefore the trench sidewalls manifest a variety of crystallographic orientations. To study this effect, large-area ($16 \mu\text{m} \times 16 \mu\text{m}$) rectangular trench transistors were fabricated such that the trench sidewalls are predominantly in the $\{100\}$ plane in one case and in the $\{110\}$ plane in the other. High-frequency $C-V$ curves from these transistors show that the gate oxide is thicker on the $\{110\}$ plane than on the $\{100\}$ plane and the flat-band voltages are different. The oxidation rate is considerably higher on the $\{110\}$ plane because the number of available Si bonds per square centimeter N_b is higher on the $\{110\}$ plane than on the $\{100\}$ plane. This is also confirmed from TEM cross sections of the trench transistor that clearly show different gate oxide thicknesses around the periphery of the trench [8].

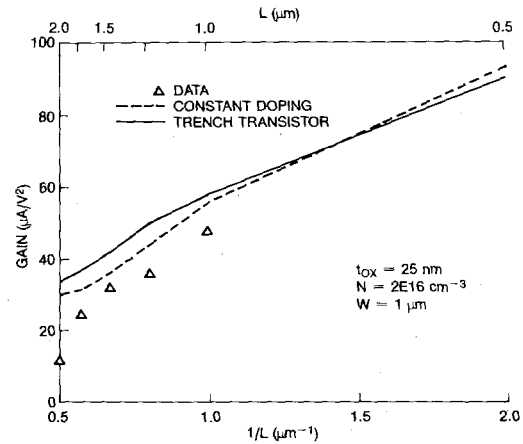


Fig. 8. Gain versus channel length for the trench transistor and the uniformly doped transistor.

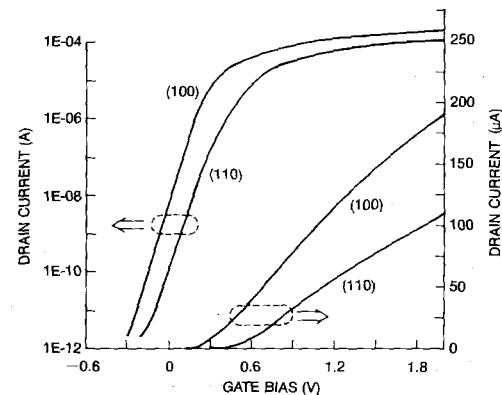


Fig. 9. Experimental linear and subthreshold characteristics for trench transistors on the $\{110\}$ and the $\{100\}$ planes.

The linear and subthreshold characteristics shown in Fig. 9 for the $\{100\}$ and $\{110\}$ planes show that the V_t and S are higher for the $\{110\}$ case, and the gain is lower mostly because of the thicker t_{ox} . It is possible to estimate the fixed oxide charge N_{ss} and fast interface state density N_{it} for the two planes from the $C-V$ curves and Fig. 9, and they are tabulated in Table I. Both N_{ss} and N_{it} are higher for the $\{110\}$ plane because of the higher Si bond density on that plane. The absolute values of N_{ss} and N_{it} are higher than for planar MOSFET's because the trenches

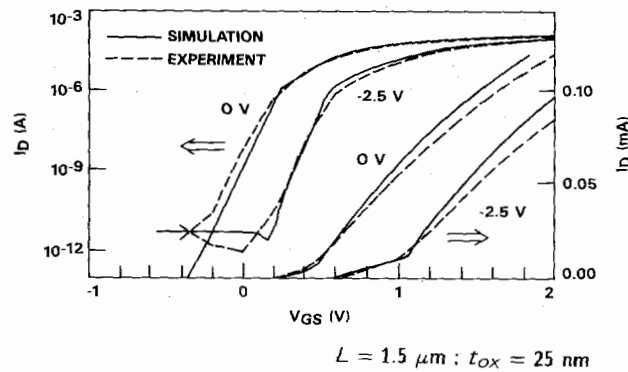


Fig. 10. Experimental and PISCES-II simulated linear and subthreshold characteristics of the trench transistor. The simulations assume two transistors $8\ \mu\text{m}/1.5\ \mu\text{m}$ and $34\ \mu\text{m}/1.5\ \mu\text{m}$ in parallel, with gate oxides of 20 and 34 nm, respectively, and the device parameters in Table I.

TABLE I
TRENCH TRANSISTOR PARAMETERS ON {100} AND {110} PLANES

PARAMETER	{100}	{110}
T_{ox} (nm)	20	34
V_t (V)	0.33	0.54
K ($\mu\text{A}/\text{V}^2$)	19.4	12.8
S (mV/dec)	78	93
N_{ss} (cm^{-2})	$3.3\text{E}10$	$1.36\text{E}11$
N_{it} ($\text{cm}^{-2}\ \text{eV}^{-1}$)	$7.0\text{E}10$	$1.61\text{E}11$
N_{bond} (cm^{-2})	$6.8\text{E}14$	$9.6\text{E}14$

TABLE II
SPICE PARAMETERS EXTRACTED FROM PISCES-II SIMULATIONS FOR 20- AND 34-nm-THICK GATE OXIDES

PARAMETER	20 nm	34 nm
C_{ox} (fF)	1.763	1.016
V_{t0} (V)	0.260	0.557
K ($\mu\text{A}/\text{V}^2$)	33.37	21.00
BE	0.332	0.611
θ (V^{-1})	0.25	0.15
N_G	0.90	0.80
KPS ($\mu\text{A}/\text{V}^2$)	0.459	0.516

are formed by reactive-ion etching, which damages the crystal surface.

The variation of t_{ox} along the perimeter of the trench explains the soft turn-on characteristics of these transistors (Fig. 2(a)). Different parts of the enclosed trench transistor have different V_t and hence turn on at different gate biases. SPICE parameters were extracted from PISCES-II simulations for 20- and 34-nm gate oxides (Table II). If the trench transistor is modeled as a parallel combination of devices with the appropriate widths and oxide thicknesses, the soft-turn behavior in the linear region can be reproduced (Fig. 10).

VII. CONCLUSION

A 2-D model based on PISCES-II simulations has been discussed to analyze the unique features of the trench transistor. It has been shown that the nonuniform doping along the channel and nonuniform gate oxides make these

devices behave differently from planar MOSFET's in certain ways. Features such as insensitivity of V_t to channel length allow these devices to be made reproducibly, allowing their application in a 4-Mbit environment.

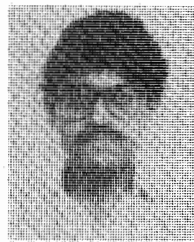
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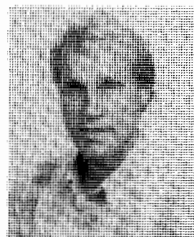
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Sanjay Banerjee (S'80-M'83) received the B.Tech. degree from the Indian Institute of Technology in 1979 and the M.S. and Ph.D. degrees from the University of Illinois in 1981 and 1983, respectively, all in electrical engineering.

He was a Member of Technical Staff at Texas Instruments in the 4Megabit Program. He is currently an Assistant Professor in the Department of Electrical and Computer Engineering at the University of Texas at Austin. His research interests are ion implantation and annealing, silicon-on-insulator and polysilicon devices, and three-dimensional IC technology.

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D. Mark Bordelon (S'82-M'83) received the B.S. degree in material science from the University of Florida, Gainesville, in 1981 and the M.S. degree in material science from the University of Texas, Austin, in 1983. His masters research involved the study of the interaction of picosecond IR radiation with crystalline silicon.

He joined Texas Instruments in 1983 and is currently a Member of the Technical Staff in the Semiconductor Process Laboratory. His present work involves process development and characterization of devices for advanced DRAM applications.